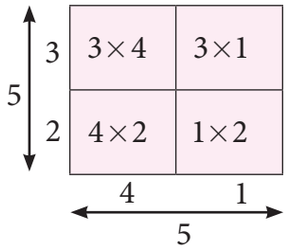
In our earlier discussion, we saw the geometrical representation of 5×3 (*Fig. 3.1*). Here we shall replace the second number 3 with 5. We get, 5×5.



We get the square number $5^2=25$, that may be represented as

$$\begin{aligned}5^2&=(3\times4)+(1\times3)+(4\times2)+(1\times2)\\&=12+3+8+2\\&=25\end{aligned}$$

Extending the same for variables in *Fig. 3.5*, the rectangular area is made into a square by making the length equal to breadth, that is $2y$ is taken as $3x$ which becomes the side of the square.

Now in *Fig. 3.6* the sides are $3x$ and area of the square is $3x \times 3x$. ...(4)

Also, each area is x^2 and since there are 9 small squares the area of whole square is $9x^2$. ...(5)

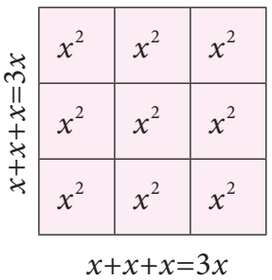


Fig. 3.6

Since both the areas [(4) and (5)] are equal, we get $3x\times3x=9x^2$.

Let us find the product of x and $(a+b)$. Look at the following figure.

In the *Fig. 3.7*, the two rectangles are combined together to form a new rectangle. Breadth of each of the rectangle are x and their lengths are a and b. So, the length and breadth of the new rectangle are $(a+b)$ and x respectively.

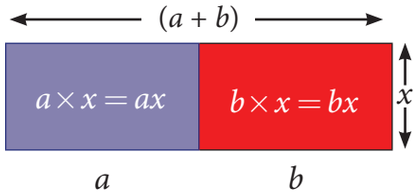


Fig. 3.7

Thus,

Area of Bigger rectangle = Area of rectangle 1 + Area of rectangle 2

Therefore, $(a+b)\times x=(a\times x)+(b\times x)$ [Distributive property]

$$=ax+bx.$$

Now, we can conclude that the multiplication of two or more monomials, as given below.

Note

Product of monomials is also a monomial.

1. If the variables are same in both the monomials then,
   - multiply the numerical co-efficient of the two monomials, separately.
   - multiply the same variable by using the product rule of exponents, that is, $a^m\times a^n=a^{m+n}$. For example, $x\times x=x^1\times x^1=x^{1+1}=x^2$.

   Also, $3x^2\times2x^3=(3\times2)(x^2\times x^3)=6\times x^{2+3}=6x^5$.
2. If the variables are different in both the monomials, then multiply them by expressing it as a product of the variables. For example, $5x\times4y=(5\times4)\times(x\times y)=20xy$.

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